

Smart EV Charging Network Simulator

C++ | Object-Oriented Programming | Data Structures | Algorithms

• Project Overview

This project simulates a real-world Smart Electric Vehicle Charging Network. As EV adoption grows rapidly across India and the world, managing charging infrastructure efficiently becomes critical. This simulator models that intelligence layer — the scheduling, routing, pricing and fault management logic that powers a real EV network.

The project is built entirely in C++ using Object-Oriented Programming principles, STL data structures, and real-world inspired algorithms.

• Key Features

1. Priority-Based Vehicle Scheduling

Vehicles are classified into three types based on real-world urgency:

- Emergency Vehicles (Ambulances, Fire Trucks, Police) — highest priority, weight = 50
- Commercial Vehicles (Delivery trucks, Ola/Uber cabs) — medium priority, weight = 30
- Private Vehicles (Personal cars) — lowest priority, weight = 20

Priority Score Formula:

```
priorityScore = typeWeight + (100 - batteryLevel) + criticalBoost
```

criticalBoost = +20 if battery level is below 20%, else 0. This ensures a dying ambulance always gets served before a fully charged private car. STL `priority_queue` automatically sorts vehicles by this score — highest score is served first.

2. Smart Routing (Greedy Algorithm)

When a new vehicle arrives, the `NetworkManager` checks all online charging stations and routes the vehicle to the station with the lowest minimum waiting time. This is a greedy algorithm — it always picks the locally optimal choice at that moment.

Time Complexity: $O(n)$ where n = number of stations.

Limitation acknowledged: In a real system, distance would also be factored in using a weighted formula: $\text{score} = \alpha \times \text{waitTime} + \beta \times \text{distance}$. This is a planned upgrade.

3. Fault Tolerance

If a charging station goes offline due to a power failure or hardware issue:

- Station is immediately marked OFFLINE
- `evacuateVehicles()` collects all vehicles from both charging slots and waiting queue
- `smartRoute()` is called for each evacuated vehicle — rerouting them to the next best online station
- The offline station returns `INT_MAX` waiting time so it is automatically excluded from routing
- Station can be manually restored via the CLI

4. Dynamic Surge Pricing

Charging rates adjust automatically based on real-time demand — similar to Uber surge pricing:

Station Condition	Charging Rate
Slots available	Rs. 10 / min (Normal)
All slots full, no queue	Rs. 15 / min (Moderate)
Queue exists	Rs. 20 / min (Surge)

Higher prices during peak demand naturally incentivizes vehicles to choose less busy stations — automatically balancing network load without any manual intervention.

5. Energy Load Balancing

Each station has a maximum power capacity (e.g., 100 kW). This power is distributed equally among all vehicles currently charging:

```
powerPerSlot = maxCapacity / activeVehicles  
chargingTime = baseTime / (powerPerSlot / maxCapacity)
```

More vehicles charging simultaneously = less power per vehicle = longer charging time. This mirrors real-world physics where power is a shared resource — like a water pipe filling multiple buckets simultaneously.

6. File Logging

Every event in the simulation is written to `ev_simulation.log` with a simulation timestamp. This includes vehicle assignments, charging completions, station failures, rerouting events and pricing changes. Similar to how production systems maintain logs for debugging and auditing.

```
[T=0 min] Vehicle 101 [Private] assigned to Station 1 | Rate: Rs.10/min  
| Cost: Rs.800  
[T=60 min] Vehicle 106 [Commercial] finished charging!  
[T=60 min] [FAULT] Station 1 went OFFLINE!
```

7. Analytics Report

At any point during or after the simulation, a full analytics report can be generated showing total vehicles served per station, revenue generated, average wait time, and overall network summary.

• System Architecture

The project follows a clean layered architecture — each class has a single clear responsibility:

File	Responsibility
Vehicle.h / .cpp	Vehicle class — priority score and charging time calculation
ChargingSlot.h / .cpp	Single charging port — time simulation and power tracking
ChargingStation.h / .cpp	Station management — slots, priority queue, pricing, load balancing

NetworkManager.h / .cpp	Network brain — smart routing, fault tolerance, analytics
Logger.h / .cpp	Event logging to file with simulation timestamps
main.cpp	Entry point — initial setup and interactive CLI menu

Sample Outputs — All Menu Options

1. Initial Setup — Vehicle Assignment

When the program starts, 7 vehicles are automatically assigned to 3 stations.

```
=====
Smart EV Charging Network Simulator
Version 2.0 - Upgraded Edition
=====

[Setup] Assigning initial vehicles...
[T=0 min] Station 1 rebalanced power: 100.000000 kW per slot (1 active)
[T=0 min] Vehicle 101 [Private] assigned to Station 1 | Rate: Rs.10/min | Est.
Cost: Rs.800 | Charging Time: 80 mins
[T=0 min] Station 1 rebalanced power: 50.000000 kW per slot (2 active)
[T=0 min] Vehicle 102 [Commercial] assigned to Station 1 | Rate: Rs.15/min | Est.
Cost: Rs.1800 | Charging Time: 120 mins
[T=0 min] All slots full at Station 1. Vehicle 103 [Emergency] added to priority
queue (score: 90) | SURGE Rate: Rs.20/min
[T=0 min] [CRITICAL] Vehicle 104 has low battery (<20%) - treating as high
priority!
[T=0 min] Station 2 rebalanced power: 80.000000 kW per slot (1 active)
[T=0 min] Vehicle 104 [Private] assigned to Station 2 | Rate: Rs.10/min | Est.
Cost: Rs.900 | Charging Time: 90 mins
[T=0 min] Station 2 rebalanced power: 40.000000 kW per slot (2 active)
[T=0 min] Vehicle 105 [Emergency] assigned to Station 2 | Rate: Rs.15/min | Est.
Cost: Rs.2250 | Charging Time: 150 mins
[T=0 min] Station 3 rebalanced power: 120.000000 kW per slot (1 active)
[T=0 min] Vehicle 106 [Commercial] assigned to Station 3 | Rate: Rs.10/min | Est.
Cost: Rs.450 | Charging Time: 45 mins
[T=0 min] Station 3 rebalanced power: 60.000000 kW per slot (2 active)
[T=0 min] Vehicle 107 [Private] assigned to Station 3 | Rate: Rs.10/min | Est.
Cost: Rs.400 | Charging Time: 40 mins
```

Option 1 — Show Network Status

Displays real-time status of all 3 stations — slots, power, pricing and queue

```

===== NETWORK STATUS [T=0 min] =====

===== Charging Station 1 =====
Max Power Capacity : 100 kW
Current Rate       : Rs.20/min [SURGE - queue exists]
Slot 1: Occupied | Vehicle 101 [Private] | Remaining: 80 mins | Power: 50 kW
Slot 2: Occupied | Vehicle 102 [Commercial] | Remaining: 120 mins | Power: 50 kW
Waiting Queue Size : 1
Min Waiting Time   : 80 mins

===== Charging Station 2 =====
Max Power Capacity : 80 kW
Current Rate       : Rs.15/min [MODERATE - all slots full]
Slot 1: Occupied | Vehicle 104 [Private] | Remaining: 90 mins | Power: 40 kW
Slot 2: Occupied | Vehicle 105 [Emergency] | Remaining: 150 mins | Power: 40 kW
Waiting Queue Size : 0
Min Waiting Time   : 90 mins

===== Charging Station 3 =====
Max Power Capacity : 120 kW
Current Rate       : Rs.10/min [NORMAL - slot available]
Slot 1: Occupied | Vehicle 106 [Commercial] | Remaining: 45 mins | Power: 60 kW
Slot 2: Occupied | Vehicle 107 [Private] | Remaining: 40 mins | Power: 60 kW
Slot 3: Free
Waiting Queue Size : 0
Min Waiting Time   : 0 mins
=====

```

Option 2 — Smart Route a New Vehicle

Added Emergency vehicle (ID: 108) at 5% battery — highest possible priority score of 165. System automatically routed it to Station 3 which had 0 mins waiting time

```

Enter Vehicle ID: 108
Enter Type (Private / Commercial / Emergency): Emergency
Enter Battery Level (0-100): 5

Vehicle ID      : 108
Type            : Emergency
Battery Level   : 5%
Priority Score   : 165
Charging Time   : 95 mins
[!] CRITICAL BATTERY - High Priority!

[T=0 min] Smart routing Vehicle 108 [Emergency] score=165
[T=0 min] --> Routed to Station 3 (wait: 0 mins)
[T=0 min] [CRITICAL] Vehicle 108 has low battery (<20%) - treating as high
priority!
[T=0 min] Station 3 rebalanced power: 40.000000 kW per slot (3 active)
[T=0 min] Vehicle 108 [Emergency] assigned to Station 3 | Rate: Rs.15/min | Est.
Cost: Rs.4275 | Charging Time: 285 mins

```

Option 3 — Simulate Time (Tick 60 Minutes)

Advanced simulation by 60 minutes. Vehicles 106 and 107 finished charging. Station 3 power rebalanced — Vehicle 108 now gets full 120 kW.

```
[T=60 min] --- Advancing simulation by 60 min(s) ---
[T=60 min] Station 1 rebalanced power: 50.000000 kW per slot (2 active)
[T=60 min] Station 2 rebalanced power: 40.000000 kW per slot (2 active)
[DONE] Vehicle 106 [Commercial] finished charging!
[DONE] Vehicle 107 [Private] finished charging!
[T=60 min] Station 3 rebalanced power: 120.000000 kW per slot (1 active)

===== NETWORK STATUS [T=60 min] =====
===== Charging Station 1 =====
  Slot 1: Occupied | Vehicle 101 [Private] | Remaining: 20 mins | Power: 50 kW
  Slot 2: Occupied | Vehicle 102 [Commercial] | Remaining: 60 mins | Power: 50 kW
  Waiting Queue Size : 1
  Min Waiting Time   : 20 mins

===== Charging Station 2 =====
  Slot 1: Occupied | Vehicle 104 [Private] | Remaining: 30 mins | Power: 40 kW
  Slot 2: Occupied | Vehicle 105 [Emergency] | Remaining: 90 mins | Power: 40 kW
  Waiting Queue Size : 0
  Min Waiting Time   : 30 mins

===== Charging Station 3 =====
  Slot 1: Free
  Slot 2: Free
  Slot 3: Occupied | Vehicle 108 [Emergency] | Remaining: 225 mins | Power: 120 kW
  Waiting Queue Size : 0
  Min Waiting Time   : 0 mins
```

Option 4 — Simulate Station Failure

Station 1 went offline. All 3 vehicles (101, 102 from slots + 103 from queue) were automatically evacuated and rerouted. Station 1 shows [OFFLINE] and INT_MAX waiting time so it is excluded from routing.

```
Enter Station ID to simulate failure: 1
```

```
[T=60 min] [FAULT] Station 1 went OFFLINE!
```

```
[FAULT TOLERANCE] Rerouting 3 vehicle(s) from Station 1...
```

```
[T=60 min] Smart routing Vehicle 101 [Private] score=100
```

```
[T=60 min] --> Routed to Station 3 (wait: 0 mins)
```

```
[T=60 min] Station 3 rebalanced power: 60.000000 kW per slot (2 active)
```

```
[T=60 min] Vehicle 101 [Private] assigned to Station 3 | Rate: Rs.10/min | Est.  
Cost: Rs.1600 | Charging Time: 160 mins
```

```
[T=60 min] Smart routing Vehicle 102 [Commercial] score=90
```

```
[T=60 min] --> Routed to Station 3 (wait: 0 mins)
```

```
[T=60 min] Station 3 rebalanced power: 40.000000 kW per slot (3 active)
```

```
[T=60 min] Vehicle 102 [Commercial] assigned to Station 3 | Rate: Rs.15/min | Est.  
Cost: Rs.5400 | Charging Time: 360 mins
```

```
[T=60 min] Smart routing Vehicle 103 [Emergency] score=90
```

```
[T=60 min] --> Routed to Station 2 (wait: 30 mins)
```

```
[T=60 min] All slots full at Station 2. Vehicle 103 [Emergency] added to priority  
queue (score: 90) | SURGE Rate: Rs.20/min
```

```
===== Charging Station 1 [OFFLINE] =====
```

```
Slot 1: Free
```

```
Slot 2: Free
```

```
Waiting Queue Size : 0
```

```
Min Waiting Time : 2147483647 mins [INT_MAX - excluded from routing]
```

```
===== Charging Station 2 =====
```

```
Current Rate : Rs.20/min [SURGE - queue now exists]
```

```
Slot 1: Occupied | Vehicle 104 [Private] | Remaining: 30 mins | Power: 40 kW
```

```
Slot 2: Occupied | Vehicle 105 [Emergency] | Remaining: 90 mins | Power: 40 kW
```

```
Waiting Queue Size : 1
```

```
Min Waiting Time : 30 mins
```

```
===== Charging Station 3 =====
```

```
Current Rate : Rs.15/min [MODERATE]
```

```
Slot 1: Occupied | Vehicle 101 [Private] | Remaining: 160 mins | Power: 40 kW
```

```
Slot 2: Occupied | Vehicle 102 [Commercial] | Remaining: 360 mins | Power: 40 kW
```

```
Slot 3: Occupied | Vehicle 108 [Emergency] | Remaining: 225 mins | Power: 40 kW
```

```
Waiting Queue Size : 0
```

```
Min Waiting Time : 160 mins
```

Option 5 — Restore Station

Station 1 restored back to ONLINE — ready to accept new vehicles.

```
Enter Station ID to restore: 1
```

```
[T=60 min] [RECOVERY] Station 1 is back ONLINE!
```

```
Station 1 restored successfully!
```

Option 6 — View Analytics Report

Full analytics report showing revenue per station and overall network summary. Station 3 served 2 vehicles (106 and 107) that completed charging during the 60-minute simulation.

```
===== NETWORK ANALYTICS REPORT =====
```

```
--- Station 1 Analytics ---
```

```
Total Vehicles Served : 0
```

```
Total Revenue      : Rs.2600
```

```
Avg Wait Time      : 0 mins
```

```
--- Station 2 Analytics ---
```

```
Total Vehicles Served : 0
```

```
Total Revenue      : Rs.3150
```

```
Avg Wait Time      : 0 mins
```

```
--- Station 3 Analytics ---
```

```
Total Vehicles Served : 2
```

```
Total Revenue      : Rs.12125
```

```
Avg Wait Time      : 0 mins
```

```
===== OVERALL SUMMARY =====
```

```
Total Vehicles Served : 2
```

```
Total Network Revenue : Rs.17875
```

```
Simulation Time     : 60 mins
```

```
=====
```

Option 7 — Exit

On exit, final analytics report is displayed and `ev_simulation.log` file is saved permanently to disk.

```
Exiting simulator.
```

```
Log saved to: ev_simulation.log
```

```
[Final analytics report displayed here]
```

ev_simulation.log

```
ev_simulation.log
1  =====
2  | Smart EV Network Simulator - Log
3  =====
4  [T=0 min] Station 1 rebalanced power: 100.000000 kW per slot (1 active)
5  [T=0 min] Vehicle 101 [Private] assigned to Station 1 | Rate: Rs.10/min | Est. Cost: Rs.800 | Charging Time: 80 mins
6  [T=0 min] Station 1 rebalanced power: 50.000000 kW per slot (2 active)
7  [T=0 min] Vehicle 102 [Commercial] assigned to Station 1 | Rate: Rs.15/min | Est. Cost: Rs.1800 | Charging Time: 120 mins
8  [T=0 min] All slots full at Station 1. Vehicle 103 [Emergency] added to priority queue (score: 90) | SURGE Rate: Rs.20/min
9  [T=0 min] [CRITICAL] Vehicle 104 has low battery (<20%) - treating as high priority!
10 [T=0 min] Station 2 rebalanced power: 80.000000 kW per slot (1 active)
11 [T=0 min] Vehicle 104 [Private] assigned to Station 2 | Rate: Rs.10/min | Est. Cost: Rs.900 | Charging Time: 90 mins
12 [T=0 min] Station 2 rebalanced power: 40.000000 kW per slot (2 active)
13 [T=0 min] Vehicle 105 [Emergency] assigned to Station 2 | Rate: Rs.15/min | Est. Cost: Rs.2250 | Charging Time: 150 mins
14 [T=0 min] Station 3 rebalanced power: 120.000000 kW per slot (1 active)
15 [T=0 min] Vehicle 106 [Commercial] assigned to Station 3 | Rate: Rs.10/min | Est. Cost: Rs.450 | Charging Time: 45 mins
16 [T=0 min] Station 3 rebalanced power: 60.000000 kW per slot (2 active)
17 [T=0 min] Vehicle 107 [Private] assigned to Station 3 | Rate: Rs.10/min | Est. Cost: Rs.400 | Charging Time: 40 mins
18 [T=0 min] Smart routing Vehicle 108 [Emergency] score=165
19 [T=0 min] --> Routed to Station 3 (wait: 0 mins)
20 [T=0 min] [CRITICAL] Vehicle 108 has low battery (<20%) - treating as high priority!
21 [T=0 min] Station 3 rebalanced power: 40.000000 kW per slot (3 active)
22 [T=0 min] Vehicle 108 [Emergency] assigned to Station 3 | Rate: Rs.15/min | Est. Cost: Rs.4275 | Charging Time: 285 mins
23 [T=60 min] --- Advancing simulation by 60 min(s) ---
24 [T=60 min] Station 1 rebalanced power: 50.000000 kW per slot (2 active)
25 [T=60 min] Station 2 rebalanced power: 40.000000 kW per slot (2 active)
26 [T=60 min] Station 3 rebalanced power: 120.000000 kW per slot (1 active)

27 [T=60 min] [FAULT] Station 1 went OFFLINE!
28 [T=60 min] Smart routing Vehicle 101 [Private] score=100
29 [T=60 min] --> Routed to Station 3 (wait: 0 mins)
30 [T=60 min] Station 3 rebalanced power: 60.000000 kW per slot (2 active)
31 [T=60 min] Vehicle 101 [Private] assigned to Station 3 | Rate: Rs.10/min | Est. Cost: Rs.1600 | Charging Time: 160 mins
32 [T=60 min] Smart routing Vehicle 102 [Commercial] score=90
33 [T=60 min] --> Routed to Station 3 (wait: 0 mins)
34 [T=60 min] Station 3 rebalanced power: 40.000000 kW per slot (3 active)
35 [T=60 min] Vehicle 102 [Commercial] assigned to Station 3 | Rate: Rs.15/min | Est. Cost: Rs.5400 | Charging Time: 360 mins
36 [T=60 min] Smart routing Vehicle 103 [Emergency] score=90
37 [T=60 min] --> Routed to Station 2 (wait: 30 mins)
38 [T=60 min] All slots full at Station 2. Vehicle 103 [Emergency] added to priority queue (score: 90) | SURGE Rate: Rs.20/min
39 [T=60 min] [RECOVERY] Station 1 is back ONLINE!
40
```

• CS Concepts Demonstrated

Concept	Where Used
Object-Oriented Programming	All classes — encapsulation, operator overloading
STL priority_queue	Priority-based vehicle scheduling in ChargingStation
STL vector	Dynamic slot and station management
Greedy Algorithm	Smart routing — O(n) minimum wait selection
Operator Overloading	operator< in Vehicle for priority_queue sorting
Resource Management	Energy load balancing across active vehicles
Fault Tolerance	Station failure detection and automatic rerouting

File I/O (ofstream)	Event logging with timestamps
Default Parameters	tick(int timeUnits = 1)
Pointer Sharing	Logger* shared across all stations

• How to Compile and Run

Requirements

- G++ compiler with C++17 support

Compile

```
g++ -std=c++17 -o ev_simulator main.cpp Vehicle.cpp ChargingSlot.cpp
ChargingStation.cpp NetworkManager.cpp Logger.cpp
```

Run

```
.\ev_simulator.exe      (Windows)
./ev_simulator          (Linux / Mac)
```

• Interactive CLI Menu

Option	Action
1	Show Network Status — view all stations, slots, queue and pricing
2	Smart Route a New Vehicle — enter vehicle details and auto-route
3	Simulate Time (Tick) — advance simulation by N minutes
4	Simulate Station Failure — take a station offline and watch rerouting
5	Restore Station — bring an offline station back online
6	View Analytics Report — revenue, vehicles served, network summary
7	Exit — saves log and displays final analytics

Known Limitations & Future Improvements

- **Distance-based routing not implemented** — charging requests are currently assigned based on station availability. A future enhancement would use a weighted scoring formula such as $score = \alpha \times waitTime + \beta \times distance$ for smarter station selection.
- **Billing is estimated upfront**- the simulation calculates charging cost before the session begins. A production system would meter actual energy consumption and generate the final bill after charging is completed.
- **Single-threaded** — stations run sequentially, not truly in parallel. Real system would use multithreading.

- **No data persistence** — application state exists only during runtime and is lost when the program terminates. Integrating a database would enable long-term storage of station, vehicle, and charging-session data.

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